

## EXPONENTIAL DISTRIBUTION

Q1: The waiting time for a bus is exponentially distributed with a mean of 10 minutes. Find the probability that a passenger waits more than 15 minutes.

Q2: The lifetime of a certain electronic component follows an exponential distribution with a mean life of 500 hours. What is the probability that a randomly chosen component lasts less than 300 hours?

Q3: Calls arrive at a call centre at an average rate of 6 per hour. Find the probability that the time between two consecutive calls is between 5 and 15 minutes.

Q4: The life of a light bulb is exponentially distributed with a mean of 800 hours. What is the probability that a bulb lasts more than 1200 hours?

Q5: If the average time between earthquakes in a region is 50 years, find the probability that the next earthquake occurs within the next 30 years.

Q6: The time between arrivals at a supermarket checkout is exponential with a mean 2 minutes. Given that no customer has arrived in the last 3 minutes, what is the probability that the next customer arrives within the next 1 minute?

Q7: A radioactive particle emits alpha particles at an average rate of 2 per second. Find the probability that the time between emissions exceeds 1 second.

Q8: The time to failure of a hard drive is exponential, with a mean 1000 hours. Find the probability that it fails between 500 and 1500 hours.

Q9: If the time between server requests is exponentially distributed with a mean 0.5 seconds, find the probability that the next request arrives in less than 0.2 seconds.

Q10: If the average time between accidents at a factory is 8 days, find the probability that the next accident occurs after 12 days.

1) 0.2231

2) 0.4512

3) 0.3834

4) 0.2231

5) 0.4512

6) 0.3935

7) 0.1353

8) 0.3834

9) 0.3297

10) 0.2231

MCQS

1: The waiting time for a bus is exponentially distributed with a mean of 8 minutes. What is the probability that a passenger waits more than 10 minutes?

- A) **0.2865**
- B) 0.7135
- C) 0.4500
- D) 0.5500

2: The time between arrivals at a service centre is exponential with mean 3 minutes. What is the probability that the time between two arrivals is less than 2 minutes?

- A) **0.4866**
- B) 0.5134
- C) 0.6321
- D) 0.3679

3: A machine fails at a rate of 0.2 failures per hour (exponential). What is the probability that the machine lasts more than 5 hours?

- A) **0.3679**
- B) 0.6321
- C) 0.8187
- D) 0.1813

4: Which of the following is TRUE for an exponential random variable?

- A) Variance = Mean
- B) Variance = Mean<sup>2</sup>
- C) Variance = 1/Mean
- D) **Variance = (Mean)<sup>2</sup>**

5: If  $X \sim \text{Exp}(\beta)$ , What is the mean of  $X$ ?

- A)  **$\beta$**
- B)  $1/\beta$
- C)  $1/\beta^2$
- D)  $\beta^2$